

Autonomous Racing Navigation: Experimental Results Summary

SONIC Project

May–June 2026

Abstract

This document consolidates lap-time and tracking metrics from all navigation experiments conducted on the AutoCar simulator. Experiments progress in three stages: (i) algorithm selection on the *circle circuit* (Stanley, MPC, Pure Pursuit), (ii) Pure Pursuit tuning on the *F1 circuit* (Albert Park), and (iii) the same F1 circuit with the Pure Pursuit + LiDAR navigation stack, including robustness sweeps over perception latency and odometry noise. All raw data are archived under **results/**.

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1 Master Comparison Table

Table 4 consolidates **all** recorded runs in a single view. Lap times from manual sessions report $T_{\min}$ (best lap) with mean $\pm$ std in parentheses; benchmark sessions report post-warmup median T_{med} . Best result per stage is marked **bold**. Rows marked TIMEOUT did not complete a measured lap within the allotted time.

Table 1: Stage I — Algorithm selection on the circle circuit (2026-05-25).

#	Algorithm	Line	$T_{\min}$ (s)	$\bar{T} \pm \sigma$ (s)	$\bar{v}$ (m/s)
1	Stanley	centerline	190.1	191.1 $\pm$ 0.9	3.42
2	Stanley	racing	172.6	173.1 $\pm$ 0.7	3.63
3	MPC	centerline	108.9	111.1 $\pm$ 2.2	5.89
4	MPC	racing	108.5	110.4 $\pm$ 1.8	5.71
5	Pure Pursuit	centerline	99.9	105.4 $\pm$ 4.8	6.23
6	Pure Pursuit	racing	92.5	95.0 $\pm$ 2.2	6.63

Table 2: Stage II — Pure Pursuit on the F1 circuit (2026-06-08).

#	Line	Profile	L (ms)	σ (m)	T_{med} (s)	$\bar{v}$ (m/s)	ε_{rms} (m)	$\dot{\delta}_{\text{max}}$ (rad/s)
7	centerline	baseline	0	0	174.6	3.52	0.768	12.46
8	racing	racing	0	0	89.5	6.41	0.295	5.07
9	racing	finetuned (conservative)	0	0	86.7	6.61	0.351	1.33
10	racing	finetuned (aggressive)	0	0	TIMEOUT	—	—	—
11	racing	finetuned_perturbed	200	0.1	98.7	18.55 [†]	0.619	3.00

Conservative (f1_finetuned): cruise_velocity=8.0 m/s, max_lateral_accel=5.5 m/s². *Aggressive* (tuning rejects, docs/rapport_optimisation_f1.md): cruise_velocity 9–10 m/s and/or max_lateral_accel=6.5 m/s² $\rightarrow$ crash or off-track, no lap-time gain; PP tracking ceilings at $\sim$ 8 m/s.

Table 3: Stage III — Pure Pursuit + LiDAR on the F1 circuit (2026-06-10).

#	Profile	L (ms)	σ (m)	T_{med} (s)	$\bar{v}$ (m/s)	ε_{rms} (m)	$\dot{\delta}_{\text{max}}$ (rad/s)	Off
12	baseline	0	0	125.6	4.17	0.071	0.80	0
13	finetuned	0	0	115.1	4.71	0.108	0.60	0
14	10_n005	0	0.05	TIMEOUT	—	—	—	—
15	10_n01	0	0.10	139.6	6.00	0.155	0.60	0
16	1200_n0	200	0	TIMEOUT	—	—	—	—
17	1200_n005	200	0.05	250.9	3.32	0.367	0.60	0
18	1200_n01	200	0.10	501.0	3.84	0.143	0.60	0
19	1500_n0	500	0	TIMEOUT	—	—	—	—
20	1500_n005	500	0.05	TIMEOUT	—	—	—	—
21	1500_n01	500	0.10	TIMEOUT	—	—	—	—

Source: results/benchmark_2026-06-10T10-52-32/summary.csv; $\dot{\delta}_{\text{max}}$ from post-warmup lap 2 (steering_rate_max in session lap_times.csv).

Table 4: Master comparison — best lap time per stage (Tables 1–3).

Stage	Track	Stack	Best lap (s)
I	Circle circuit	Stanley / MPC / Pure Pursuit	92.5
II	F1 circuit	Pure Pursuit	86.7
III	F1 circuit	Pure Pursuit + LiDAR	115.1

[†]Stage II perturbed run: inflated path distance (1831 m vs. ≈ 573 m) under localisation noise; $\bar{v}$ is not comparable to nominal runs. Stage I lap column format: $T_{\min} (\bar{T} \pm \sigma)$. Stages II–III: T_{med} and δ_{max} from post-warmup lap 2 (warmup lap 1 excluded from summary).

2 Experimental Setup

2.1 Tracks and navigation stacks

- **Circle circuit** — oval track used for controller comparison (`results/stanley_2026-05-25T13-55-37`, etc.). Nominal lap length from waypoint polylines (`waypoints.csv` / `waypoints_racing.csv`): centerline ≈ 651 m, racing line ≈ 627 m.
- **F1 circuit** — Albert Park layout (scaled for simulation), used in both Stage II and Stage III (`results/benchmark_2026-06-08T10-52-32`, `results/benchmark_2026-06-10T10-52-32`). Nominal lap length from precomputed waypoint polylines (`waypoints_f1.csv` / `waypoints_f1_racing.csv`; Stage II only): centerline ≈ 596 m, racing line ≈ 568 m (real Melbourne GP circuit: ≈ 5.3 km). Stage II employs the standard Pure Pursuit stack with these CSV paths loaded at launch. Stage III employs Pure Pursuit + LiDAR: lap 1 explores via LiDAR; centerline and racing line are *computed online* from the SLAM map and stored in memory (not read from CSV). The benchmark config label `f1_circuit_fenced` refers to this LiDAR-enabled setup on the same circuit geometry, not a separate track.

2.2 Trajectory modes

- **Stages I–II (open-loop stacks)** — `centerline` and `racing` refer to precomputed waypoint files (`waypoints*.csv`) loaded at launch.
- **Stage III (Pure Pursuit + LiDAR)** — the `line` launch flag selects the same nominal mode names, but trajectories are *computed at runtime* and held in memory: lap 1 builds a local centerline from LiDAR (Follow-the-Gap) and records the exploration path into the SLAM map; at lap 2 the `global_planner_lidar` node derives a closed-loop centerline from that path, optimises a min-curvature racing line, smooths it, and caches the result for subsequent laps (typical lap-2+ distance ≈ 520 – 540 m in our runs, vs. ≈ 568 m for the offline `waypoints_f1_racing.csv`).

2.3 Metrics

- **Lap time** T_{lap} (s) — elapsed time between consecutive start/finish crossings.
- **Average speed** $\bar{v}$ (m/s) and **peak speed** v_{max} (m/s).
- **Path distance** d (m) — integrated odometry over the lap.
- **Lateral error RMS** ε_{rms} (m) and **peak lateral error** ε_{max} (m) — benchmark runs only.
- **Peak steering rate** $\dot{\delta}_{\text{max}}$ (rad/s) — maximum $|\dot{\delta}/\text{dt}|$ per lap (proxy for steering oscillation).
- **Off-track events** — count of boundary violations (benchmark runs only).

2.4 Local planner curvature speed limit

The local planner (`local_planner`) publishes `/autocar/target_velocity` at `update_frequency` (10 Hz). It caps speed from the peak curvature κ_{peak} on the cubic-spline path ahead of the front axle. Let i be the closest-path index and $N_{\text{cl}} = \text{curvature_lookahead}$ (path samples, not metres; at 10 Hz sampling $ds \approx 0.1$ m per point, so $N_{\text{cl}} = 110 \Rightarrow \approx 11$ m arc length). The planner takes the maximum of a moving-average filtered $|\kappa|$ over $[i, i + N_{\text{cl}})$ (`curvature_smooth_window`, default 5):

$$\kappa_{\text{peak}} = \max_{j \in [i, i + N_{\text{cl}})} \text{MA}(|\kappa_j|) \quad (1)$$

Centripetal acceleration $a_{\text{lat}} = v^2 |\kappa|$ yields the curvature-limited speed

$$v_{\text{curve}} = \sqrt{\frac{a_{\text{lat}, \text{max}}}{\max(|\kappa_{\text{peak}}|, \kappa_{\text{min}})}} \quad (2)$$

with $a_{\text{lat,max}} = \text{max_lateral_accel}$ and $\kappa_{\text{min}} = \text{min_curvature}$ (prevents $v_{\text{curve}} \rightarrow \infty$ on straights). The cruise cap depends on `nav_mode`: $v_{\text{cruise}} = \text{exploration_velocity}$ (lap 1) or `cruise_velocity` (lap 2+):

$$v_{\text{target}} = \min(v_{\text{cruise}}, v_{\text{curve}}) \quad (3)$$

The published target is a ramped value v_{pub} that approaches v_{target} asymmetrically each tick ($\Delta t = 1/\text{update_frequency}$) via `accel_rate` / `decel_rate`; the path tracker follows v_{pub} (with optional error-based scaling). Example (Stage II finetuned, $a_{\text{lat,max}} = 5.5 \text{ m/s}^2$, $|\kappa_{\text{peak}}| = 0.10 \text{ m}^{-1}$): $v_{\text{curve}} \approx \sqrt{5.5/0.10} \approx 7.4 \text{ m/s}$; with $v_{\text{cruise}} = 8.0 \text{ m/s}$, Eq. 3 gives $v_{\text{target}} = 7.4 \text{ m/s}$.

2.5 Pure Pursuit lookahead distance

The path tracker (`path_tracker`) sets the arc-length lookahead L_d^{final} (m) each control cycle from the *measured* speed v and the peak path curvature κ_{peak} over the segment ahead of the closest point (`lookahead_curv_window`, default 30 path samples):

$$L_d = \text{clip}(g_{\text{ld}} v + L_{\text{min}}, L_{\text{min}}, L_{\text{max}}) \quad (4)$$

$$\text{scale}(\kappa_{\text{peak}}) = \frac{1}{1 + (|\kappa_{\text{peak}}|/\kappa_{\text{soft}})^2} \quad (5)$$

$$L_d^{\text{final}} = L_d + L_{\text{curv_extra}} \text{scale}(\kappa_{\text{peak}}) \quad (6)$$

Mapping to ROS parameters: $g_{\text{ld}} = \text{lookahead_gain}$, $L_{\text{min}} / L_{\text{max}} = \text{lookahead_min} / \text{lookahead_max}$, $L_{\text{curv_extra}} = \text{lookahead_curv_extra}$ (0 m disables the curvature-adaptive term; baseline Stage II/III), $\kappa_{\text{soft}} = \text{lookahead_curv_soft}$. On straights $\text{scale} \approx 1$; in sharp bends $\text{scale} \rightarrow 0$, so the extra term lengthens lookahead only on straight segments (anti-zigzag) without cutting corners. Example (Stage III finetuned, $v = 7.5 \text{ m/s}$): $g_{\text{ld}} = 0.48$, $L_{\text{min}} = 1.8 \text{ m}$, $L_{\text{max}} = 8.0 \text{ m} \Rightarrow L_d \approx 5.4 \text{ m}$ (Eq. 4); with $L_{\text{curv_extra}} = 4.0 \text{ m}$ on a straight, $L_d^{\text{final}} \approx 9.4 \text{ m}$ (Eq. 6).

2.6 Recording conventions

Manual launches write per-lap records to `results/<stack>_<run_id>/lap_times.csv` (15 columns when extended metrics are enabled). Batch benchmarks (`scripts/benchmark.py`) aggregate results into `summary.csv` with one warmup lap excluded from reported statistics (`warmup_laps = 1`). Runs that did not complete a measured lap within the allotted time are marked **timeout**.

3 Algorithm Selection — Circle Circuit

Three path-tracking stacks were evaluated on the circle circuit under both centerline and racing trajectories. Sessions were recorded on 2026-05-25. Pure Pursuit achieved the lowest lap times and was retained for subsequent experiments.

3.1 Summary comparison

Table 5 reports aggregate statistics over all recorded laps. Racing-line trajectories shorten the path ($\approx 630 \text{ m}$ vs. $\approx 653 \text{ m}$) and generally reduce lap time.

Table 5: Circle-circuit algorithm comparison (2026-05-25). Best lap time per trajectory mode in **bold**.

Stack	Line	n	$\bar{T}_{\text{lap}}$	σ_T	T_{min} (s)	T_{max}	$\bar{v}$ (m/s)	v_{max}
Stanley	centerline	3	191.1	0.9	190.1	191.8	3.42	5.85
Stanley	racing	2	173.1	0.7	172.6	173.6	3.63	5.91
MPC	centerline	3	111.1	2.2	108.9	113.2	5.89	7.58
MPC	racing	5	110.4	1.8	108.5	113.0	5.71	7.56
Pure Pursuit	centerline	3	105.4	4.8	99.9	109.0	6.23	7.60
Pure Pursuit	racing	3	95.0	2.2	92.5	96.8	6.63	7.67

3.2 Racing-line speedup over centerline

Table 6: Racing-line improvement relative to centerline on the circle circuit.

Stack	$\bar{T}_{\text{center}}$ (s)	$\bar{T}_{\text{racing}}$ (s)	ΔT (%)
Stanley	191.1	173.1	−9.4
MPC	111.1	110.4	−0.6
Pure Pursuit	105.4	95.0	− 9.8

3.3 Per-lap records

3.3.1 Stanley

Table 7: Stanley — circle circuit. Source: `results/stanley_2026-05-25T13-55-37` (centerline), `results/stanley_racing_2026-05-25T13-48-10` (racing).

Line	Lap	T_{lap} (s)	$\bar{v}$ (m/s)	v_{max} (m/s)	d (m)
centerline	1	191.3	3.41	5.83	652.5
centerline	2	190.1	3.43	5.85	652.5
centerline	3	191.8	3.40	5.84	652.8
racing	1	172.6	3.64	5.83	628.3
racing	2	173.6	3.62	5.91	628.6

3.3.2 MPC

Table 8: MPC — circle circuit. Source: `results/mpc_2026-05-25T16-46-36` (centerline), `results/mpc_racing_2026-05-25T16-56-47` (racing).

Line	Lap	T_{lap} (s)	$\bar{v}$ (m/s)	v_{max} (m/s)	d (m)
centerline	1	113.2	5.78	7.46	653.9
centerline	2	111.1	5.89	7.58	654.0
centerline	3	108.9	6.00	7.41	653.6
racing	1	113.0	5.58	7.31	630.6
racing	2	108.5	5.80	7.51	629.7
racing	3	110.4	5.71	7.45	630.2
racing	4	110.8	5.69	7.49	630.3
racing	5	109.1	5.78	7.56	630.1

3.3.3 Pure Pursuit (selected algorithm)

Table 9: Pure Pursuit — circle circuit. Source: `results/pure_pursuit_2026-05-25T16-28-02` (centerline), `results/pure_pursuit_racing_2026-05-25T16-34-12` (racing).

Line	Lap	T_{lap} (s)	$\bar{v}$ (m/s)	v_{max} (m/s)	d (m)
centerline	1	99.9	6.55	7.60	654.1
centerline	2	107.2	6.12	7.56	656.4
centerline	3	109.0	6.03	7.58	657.2
racing	1	96.8	6.51	7.52	629.7
racing	2	95.7	6.58	7.50	629.5
racing	3	92.5	6.80	7.67	629.0

3.4 Selection outcome

Pure Pursuit delivered the **fastest best lap** (92.5s, racing line) and the **largest racing-line gain** (−9.8%). Stanley was approximately 80s slower per lap; MPC was competitive on centerline but did not outperform Pure Pursuit. Pure Pursuit was therefore chosen for all F1-circuit experiments.

4 F1 Circuit — Pure Pursuit Progression

Four configurations were benchmarked on the Albert Park F1 circuit on 2026-06-08 ([results/benchmark_2026-06-08T19-47-00](#)). Each run completed two laps with one warmup lap excluded from the summary.

Table 10: Pure Pursuit on the F1 circuit: baseline, racing line, parameter finetuning, and finetuned stack under injected latency/noise. Source: [results/benchmark_2026-06-08T19-47-00/summary.csv](#).

Profile	Line	Lat. (ms)	Odom. σ	T_{med} (s)	$\bar{v}$ (m/s)	v_{max} (m/s)	ε_{rms} (m)	$\dot{\delta}_{\text{max}}$ (rad/s)
baseline	centerline	0	0.0	174.6	3.52	7.54	0.768	12.46
racing	racing	0	0.0	89.5	6.41	7.63	0.295	5.07
finetuned	racing	0	0.0	86.7	6.61	7.69	0.351	1.33
finetuned_perturbed	racing	200	0.1	98.7	18.55 [†]	7.32	0.619	3.00

[†]Elevated $\bar{v}$ reflects inflated path distance (1831 m vs. ≈ 573 m) under perturbation, not higher on-track speed.

4.1 Per-lap records

Table 11 lists both laps for each Stage II run (warmup lap 1, measured lap 2). Sources: [results/pure_pursuit_2026-06-08T19-47-00](#) (baseline), [results/pure_pursuit_racing_2026-06-08T19-53-41](#) (racing), [results/pure_pursuit_racing_2026-06-08T19-57-17](#) (finetuned), [results/pure_pursuit_racing_2026-06-08T20-00-46](#) (perturbed).

Table 11: F1 circuit — per-lap records (2026-06-08).

Profile	Lap	T_{lap} (s)	$\bar{v}$ (m/s)	v_{max} (m/s)	ε_{rms} (m)	$\dot{\delta}_{\text{max}}$ (rad/s)	Off
baseline	1	169.2	3.61	7.65	0.711	7.71	0
baseline	2	174.6	3.52	7.54	0.768	12.46	0
racing	1	95.6	6.04	7.66	0.561	14.79	0
racing	2	89.5	6.41	7.63	0.295	5.07	0
finetuned	1	87.8	6.52	7.74	0.359	0.80	0
finetuned	2	86.7	6.61	7.69	0.351	1.33	0
perturbed	1	99.9	18.49	7.32	0.677	3.71	0
perturbed	2	98.7	18.55	7.32	0.619	3.00	0

4.2 Finetuned profile — parameter changes (Stage II)

Navigation overrides are archived under [results/benchmark_2026-06-08T19-47-00/nav_overrides/](#) (materialised from [scripts/configs/f1_circuit.yaml](#)). The open-loop Pure Pursuit stack uses precomputed waypoints; the **finetuned** profile (**f1_finnetuned**) builds on **racing** (**f1_racing**) with one additional path-tracker change (Table 12). Stage III finetuned parameters (LiDAR stack) are documented separately in Section 5.2 — the same profile name, but a different tuning set.

Table 12: Stage II parameter deltas for the F1 finetuned profile (canonical: `f1_circuit.yaml`; archived as `run_001 f1_baseline`, `run_002 f1_racing`, `run_003 f1_finnetuned`).

Parameter	Baseline	Racing	Finetuned	Role
Trajectory (<code>line</code>)	centerline	racing	racing	Switch to the min-curvature QP racing line (<code>waypoints_f1_racing.csv</code>), smoothed offline (<code>smooth_racing_line.py</code>).
<code>curvature_lookahead</code>	140	60	60	Shorter bend-detection horizon on the twisty F1 layout (Eqs. 1–3): brake later instead of treating the whole circuit as one continuous corner.
<code>heading_soft</code>	0.6	1.2	1.2	Relax heading-error penalty so throttle is not crushed in corners (baseline setting cut speed to $\approx 16\%$ of target).
<code>lookahead_curv_extra</code>	0	0	3.0 m	Finetuned-only: add extra lookahead on straights only; suppresses high-speed zigzag without widening lookahead in corners.
<code>lookahead_gain</code> / <code>lookahead_min</code>	0.4 / 1.5	0.3 / 1.0	0.3 / 1.0	Stack default Pure Pursuit gains shortened for F1 tight corners (L_d at 8 m/s: 4.7 m $\rightarrow$ 3.4 m).

Baseline column: stack defaults (`navigation_params.yaml`). The `f1_baseline` run also uses 0.3 / 1.0 in `f1_circuit.yaml` (F1 circuit override applied before racing-line / finetuning steps).

Incremental change (racing $\rightarrow$ finetuned). Only `lookahead_curv_extra` differs between `run_002` and `run_003`; it accounts for the 3.1% lap-time gain (89.5 s $\rightarrow$ 86.7 s) and the drop in $\dot{\delta}_{\max}$ (5.07 $\rightarrow$ 1.33 rad/s). All other navigation parameters are identical between the two profiles. `f1_finnetuned_perturbed` reuses the `f1_finnetuned` navigation block; only `latency_ms` = 200 and `odom_noise_std` = 0.1 differ.

4.3 Key observations

1. Switching from centerline to racing line on F1 cuts median lap time by 49% (174.6 s $\rightarrow$ 89.5 s).
2. Parameter finetuning yields a further 3.1% reduction (89.5 s $\rightarrow$ **86.7 s**).
3. Under 200 ms latency and odometry noise ($\sigma = 0.1$ m), the finetuned controller remains stable (zero off-track events) but lap time increases by 13.8% and path distance diverges, indicating degraded localisation.
4. Lateral tracking improves markedly with the racing line: ε_{rms} drops from 0.768 m (centerline) to 0.295 m (racing).
5. Parameter finetuning also smooths steering: peak steering rate on the measured lap falls from 5.07 rad/s (racing) to **1.33 rad/s** (finetuned), a -74% reduction in steering-rate spikes.

5 F1 Circuit — Pure Pursuit with LiDAR Stack

Stage III reuses the same F1 circuit as Stage II, but switches to the Pure Pursuit + LiDAR navigation stack, which constrains the vehicle to LiDAR-detected track boundaries. Ten configurations were evaluated on 2026-06-10 (`results/benchmark_2026-06-10T10-52-32`), sweeping perception latency $L \in \{0, 200, 500\}$ ms and odometry noise $\sigma \in \{0, 0.05, 0.1\}$ m.

5.1 Nominal conditions ($L = 0$, $\sigma = 0$)

Table 13: Pure Pursuit + LiDAR under nominal sensing on the F1 circuit (racing line).

Profile	T_{med} (s)	$\bar{v}$ (m/s)	v_{max} (m/s)	d (m)	ε_{rms} (m)	$\dot{\delta}_{\text{max}}$ (rad/s)	Off-track
baseline	125.6	4.17	5.77	523.4	0.071	0.80	0
finetuned	115.1	4.71	7.25	542.1	0.108	0.60	0

Finetuning reduces lap time by 8.4% while maintaining sub-0.11 m lateral error RMS and lowering peak steering rate from 0.80 to **0.60 rad/s** on lap 2 (-25%). Parameter changes vs. baseline are listed in Table 14.

5.2 Finetuned profile — parameter changes (Stage III)

Stage III uses the Pure Pursuit + LiDAR stack: lap 1 builds a SLAM map and extracts a smoothed racing line; lap 2+ races on that online trajectory. Overrides are archived under `results/benchmark_2026-06-10T10-52-32/nav_overrides/` (`run_000` baseline, `run_001` finetuned; source config `f1_pure_pursuit_lidar.yaml`). Unlike Stage II, finetuning here retunes *all three* navigation nodes (LiDAR global planner, local planner, path tracker).

Table 14: Stage III parameter deltas for the F1 finetuned profile (Pure Pursuit + LiDAR; baseline vs. finetuned).

Parameter	Baseline	Finetuned	Role
<i>global_planner_lidar</i>			
<code>waypoints_ahead / waypoints_behind</code>	5 / 2	6 / 1	More lookahead spline goals, fewer behind — smoother path at lap seam.
<code>waypoint_search_ahead</code>	30	35	Wider index search for stable goal selection at 7.5 m/s.
<code>centerline_step</code>	3.5	3.0 m	Denser lap-1 exploration path → better online racing line.
<code>racing_mincurv_max_offset</code>	5.0	4.5 m	Less aggressive corner cutting on the min-curvature racing line.
<code>racing_smooth_iters / racing_smooth_pre_iters</code>	10 / 4	12 / 5	Extra smoothing of the SLAM-derived racing trajectory.
<code>racing_smooth_alpha / racing_smooth_pre_alpha</code>	0.40	0.42	Slightly stronger smoothing blend on extracted waypoints.
<i>local_planner</i>			
<code>cruise_velocity / avoid_velocity</code>	6.0	7.5 m/s	Raise lap-2+ target speed; baseline was conservative for SLAM exploration.
<code>max_lateral_accel</code>	4.5	5.0	Allow slightly higher cornering speed (Eq. 2); 5.5 was too nervous on this stack.
<code>curvature_lookahead</code>	140	110	Shorten bend horizon (Eqs. 1–3; Stage II used 60, but here 60 caused lap-2 speed bobbing).
<code>curvature_smooth_window</code>	5	7	Smooth cruise-velocity transitions when curvature estimate changes.
<code>decel_rate</code>	6.0	6.5	Slightly faster braking into corners at higher cruise speed.
<i>path_tracker</i>			
<code>lookahead_gain / lookahead_min / lookahead_max</code>	0.4 / 1.5 / 6.0	0.48 / 1.8 / 8.0	Longer adaptive lookahead at 7.5 m/s ($L_d \approx 5.4$ m, Eq. 4) to cut zigzag.
<code>lookahead_curv_extra</code>	0	4.0 m	Straight-only lookahead extension (Stage II: 3.0 m; higher speed needs more).
<code>lookahead_curv_soft</code>	0.08	0.10	Ignore high-frequency spline-curvature noise when adapting lookahead.
<code>closest_search_ahead</code>	120	140	Stabilise closest-point index at higher lap speed.
<code>steering_limits</code>	0.95	0.92	Lower steering cap to suppress peak steering commands.
<code>heading_soft</code>	0.6	0.65	Mild heading-penalty relax (Stage II used 1.2; here throttle modulation did not fix zigzag).

Perturbed profiles (`finetuned_perturbed_*`). The eight sweep runs share the same navigation block as

finetuned; only `latency_ms` and `odom_noise_std` differ ($L \in \{0, 200, 500\}$ ms, $\sigma \in \{0, 0.05, 0.1\}$ m).

Unchanged on purpose (both stages). `steer_smoothing` stays at 1.0 (off): low-pass filtering amplifies oscillation via phase lag. Stage III keeps `exploration_velocity` at 3.0 m/s for a stable lap-1 SLAM pass.

5.3 Robustness sweep

Table 15: Pure Pursuit + LiDAR robustness matrix. TIMEOUT = run timed out before completing a measured lap. Source: `results/benchmark_2026-06-10T10-52-32/summary.csv`.

Profile	L (ms)	σ (m)	Status	T_{med} (s)	$\bar{v}$ (m/s)	ε_{rms} (m)	$\dot{\delta}_{\text{max}}$ (rad/s)	Off-track
finetuned_perturbed_l0_n005	0	0.05	TIMEOUT	—	—	—	—	—
finetuned_perturbed_l0_n01	0	0.10	OK	139.6	6.00	0.155	0.60	0
finetuned_perturbed_l200_n0	200	0.00	TIMEOUT	—	—	—	—	—
finetuned_perturbed_l200_n005	200	0.05	OK	250.9	3.32	0.367	0.60	0
finetuned_perturbed_l200_n01	200	0.10	OK	501.0	3.84	0.143	0.60	0
finetuned_perturbed_l500_n0	500	0.00	TIMEOUT	—	—	—	—	—
finetuned_perturbed_l500_n005	500	0.05	TIMEOUT	—	—	—	—	—
finetuned_perturbed_l500_n01	500	0.10	TIMEOUT	—	—	—	—	—

5.4 Robustness summary

- **Latency sensitivity:** all 500 ms configurations timed out. At 200 ms, only runs with $\sigma > 0$ completed; pure-latency injection ($L=200$ ms, $\sigma=0$) failed.
- **Noise-only perturbation** ($L=0$): moderate noise ($\sigma=0.05$ m) caused timeout; $\sigma=0.10$ m completed with $T_{\text{med}} = 139.6$ s (+21% vs. finetuned nominal).
- **Combined perturbation** ($L=200$ ms): lap times scale severely (250.9 s at $\sigma=0.05$ m; 501.0 s at $\sigma=0.10$ m), yet no off-track events were recorded.
- **Best LiDAR result:** finetuned profile, **115.1 s** median lap under nominal sensing.

6 Cross-Experiment Comparison

Table 4 (Section 1) summarises the best lap time per stage; Tables 1–3 list all individual runs. Direct comparison across stages is indicative only — the circle and F1 circuits differ in layout; Stages II and III share the same F1 circuit but use different navigation stacks.

Table 16: Best lap times by experimental stage (same as Table 4).

Stage	Track	Stack	Best T_{lap} (s)	Source
Algorithm selection	circle	Pure Pursuit	92.5	<code>results/pure_pursuit_racing_2</code>
F1 progression	F1 circuit	Pure Pursuit	86.7	<code>results/pure_pursuit_racing_2</code>
F1 + LiDAR	F1 circuit	Pure Pursuit + LiDAR	115.1	<code>results/benchmark_2026-06-10T</code>

7 Conclusions

1. **Algorithm:** Pure Pursuit outperformed Stanley and MPC on the circle circuit, achieving a best lap of 92.5 s on the racing line.
2. **Racing line:** switching from centerline to racing consistently reduces lap time (−9.8% on circle; −49% on F1).
3. **F1 finetuning:** parameter tuning on Pure Pursuit yields **86.7 s** on the F1 circuit (Stage II) — the overall best result — and reduces peak steering rate from 5.07 to 1.33 rad/s (−74%).
4. **LiDAR stack:** switching from Pure Pursuit to Pure Pursuit + LiDAR on the same F1 circuit increases lap time by ≈ 28 s (115.1 s vs. 86.7 s, both finetuned, racing line), but peak steering rate stays low (0.60 rad/s vs. 1.33 rad/s in Stage II).
5. **Robustness:** the LiDAR stack tolerates moderate combined perturbation without off-track events, but 500 ms latency is beyond the operational envelope; 200 ms combined with noise degrades performance substantially.

A Data Index

Table 17: Result directories referenced in this report.

Experiment	Directory	Date
Stanley, circle, centerline	results/stanley_2026-05-25T13-55-37	2026-05-25
Stanley, circle, racing	results/stanley_racing_2026-05-25T13-48-10	2026-05-25
MPC, circle, centerline	results/mpc_2026-05-25T16-46-36	2026-05-25
MPC, circle, racing	results/mpc_racing_2026-05-25T16-56-47	2026-05-25
Pure Pursuit, circle, centerline	results/pure_pursuit_2026-05-25T16-28-02	2026-05-25
Pure Pursuit, circle, racing	results/pure_pursuit_racing_2026-05-25T16-34-12	2026-05-25
PP, F1 baseline (centerline)	results/pure_pursuit_2026-06-08T19-47-06	2026-06-08
PP, F1 racing line	results/pure_pursuit_racing_2026-06-08T19-53-41	2026-06-08
PP, F1 finetuned	results/pure_pursuit_racing_2026-06-08T19-57-17	2026-06-08
PP, F1 finetuned + perturbation	results/pure_pursuit_racing_2026-06-08T20-00-46	2026-06-08
PP, F1 benchmark (aggregate)	results/benchmark_2026-06-08T19-47-00	2026-06-08
Pure Pursuit + LiDAR, F1 circuit	results/benchmark_2026-06-10T10-52-32	2026-06-10
PP + LiDAR session runs ($\times 10$)	results/benchmark_2026-06-10T10-52-32/summary.csv (run_dir)	2026-06-10